Fiscal Year:	FY 2008	**Task Last Updated:**	FY 02/16/2009
PI Name:	Weislogel, Mark M. Ph.D.		
Project Title:	Capillary Flow in Complex Geometries		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mmw@cecs.pdx.edu	**Fax:**	FY 503-725-8255
PI Organization Type:	UNIVERSITY	**Phone:**	503 725-4292
Organization Name:	Portland State University		
PI Address 1:	MCECS Research		
PI Address 2:	P.O. Box 751		
PI Web Page:	https://		
City:	Portland	**State:**	OR
Zip Code:	97207-0751	**Congressional District:**	3
Comments:	Professional affiliations for Dr. Weislogel (updated 1/2/2025): IRPI, LLC 27501 SW 95th Ave, Suite 930 Wilsonville, OR 97070-2905 and Portland State University MCECS Research, P.O. Box 751 Portland, OR 97207-0751		
Project Type:	Flight	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
Start Date:	05/04/2005	**End Date:**	05/31/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	8	**Monitoring Center:**	NASA GRC
Contact Monitor:	Rame, Enrique	**Contact Phone:**	216-433-2842
Contact Email:	Rame@grc.nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC05AA29A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Knowledge of capillary flows in complex geometries is fundamental to many space fluids systems designs. Such knowledge provides a basis to exploit passive means of containment and management for large liquid inventories aboard spacecraft. Such knowledge is also essential to prevent inadvertent destabilization and deactivation of critical systems where capillary forces can dominate fluid behavior. Complimentary but distinctly non-computationally-intensive methods to design and analysis such are in need by the aerospace community tasked to meet NASA's challenges for space exploration in the present and future. Specific progress is reported here for three fundamental and applied research projects: (1) the NRA investigation "Capillary Flow in Complex Geometries," (2) ISS Flight experiment results and analysis for the "Capillary Flow Experiment" (CFE), development of follow-on investigation CFE-2 also for ISS, and (3) progress in the development of the German-US collaborative ISS experiment "Capillary Channel Flow" (CCF) experiment. All are briefly highlighted in this summary and addressed individually in greater detail in the subsequent report.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many of the results of this work are immediately useful in the design of fluid systems aboard space craft. Certain results also inform the design and analysis methods for porous structures and media for the study of transport within porous media on earth such as ground water flows, high performance wicks for heat pipes, improved Lab-on Chip methods, and addressing flows in fuel cells. Three patents have been applied for and two systems designed and tested for application in a reduced gravity environments realting to this work.
Task Progress:	The Capillary Flow Experiments on the ISS were conducted in space beginning with Increment 9 through Increment 16, beginning August, 2004 and ending December, 2007. Both `primary' and `extra science' experiments were conducted during 19 operations performed by 7 astronauts including: M. Fincke, W. McArthur, J. Williams, S. Williams, M. Lopez-Alegria, C. Anderson, and P. Whitson. CFE consists of 6 approximately 1 to 2kg handheld experiment units designed to investigate a selection of capillary phenomena of fundamental and applied importance, such as large length scale contact line dynamics (CFE-Contact Line), critical wetting in discontinuous structures (CFE-Vane Gap), and capillary flows and passive phase separations in complex containers (CFE-Interior Corner Flow). Highly quantitative video from the simply performed flight experiments provide data helpful in benchmarking numerical methods, confirming theoretical models, and guiding new model development. In several publications, the history of the experiment is reviewed introducing the science investigated. A selection of experimental results and comparisons with both analytic and numerical predictions has been provided in the literature. A more exhaustive report is expected to publish shortly as a NASA Contractor Report. Despite the simple nature of the experiments and procedures, many of the experimental results may be practically employed to enhance the design of spacecraft engineering systems involving capillary interface dynamics. As a result of their success, follow-on experiments (CFE-2) is underway, with new test cell under fabrication and slated to fly to ISS by 2010. Progress towards a 2010 launch of the Capillary Channel Flow (CCF; M. Dreyer PI) has also been maintained. Progress in the prediction of complex capillary flows in weakly 3-D geometries has also provided several useful predictive tools for low-g fluids system design. These enable designers to quickly design and analyze optimal structures for passive fluids handling. The devices are demonstrated to perform passive phase separations and to be less sensitive to wetting conditions as were previous methods. Complex interactions of numerous interconnected pathways remain under investigation.
Bibliography Type:	Description: (Last Updated: 06/13/2025)
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